Enterprise Supplier Research Agent

Literature Summary

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Contents

1 Objective **2**

2 Core Requirements **2**

3 Proposed Architecture **2**
- 3.1 Stage 1: Input Ingestion . . . 2
- 3.2 Stage 2: Research Planning . . . 2
- 3.3 Stage 3: Web Research . . . 3
- 3.4 Stage 4: Evidence Storage . . . 3
- 3.5 Stage 5: Comparison and Reasoning . . . 3
- 3.6 Stage 6: Citation Verification . . . 3
- 3.7 Stage 7: Final Output . . . 3
- 3.8 Iteration . . . 3

4 Literature Summary **4**
- 4.1 Reasoning Formats and GSM8K Performance . . . 4
- 4.2 Key Observations from the Paper . . . 5

5 Proposed Models **5**
- 5.1 Reasoning Model Candidates . . . 5
- 5.2 DeepSeek V3.1 Size Reference . . . 5
- 5.3 Smaller Models for Colab Prototyping . . . 6

6 Browsing Layer **6**
- 6.1 Candidate Tools . . . 6
- 6.2 Blocking Reality . . . 6

7 Summary **6**

Appendix: Referenced Sources **7**

1 Objective

This document records the technical groundwork for an agentic system that:

1. Ingests supplier quotations (PDF, XLSX, CSV).
2. Researches each supplier's specifications, pricing, and reputation across the web.
3. Produces a ranked recommendation with **verifiable citations** explaining *why* a supplier was selected.
4. Runs entirely on open-source, commercially licensed components so it can be self-hosted and audited.

The system is intended for enterprise procurement use, where auditability, licensing safety, and citation traceability are mandatory.

2 Core Requirements

Requirement	Description
Research capability	Agent must browse supplier websites, catalogs, and reviews.
Quotation parsing	Extract unit prices, currencies, lead times, and terms from uploaded files.
Reasoning transparency	The model must explain *why* a supplier was chosen, with evidence.
Citation verification	Every factual claim must trace back to a real, retrieved source.
Anti-blocking	The browsing layer must reduce detection and degrade gracefully.
Commercial licensing	All components must permit unrestricted enterprise use.
Self-hostable	Must run on internal infrastructure for data control and audit.

Table 1: Core system requirements.

3 Proposed Architecture

The system operates as a sequential pipeline. Each stage produces an artifact that feeds the next, and the pipeline can loop back for iterative research before producing a final recommendation.

3.1 Stage 1: Input Ingestion

The user uploads a supplier quotation (PDF, XLSX, or CSV) along with specification requirements.

- **Input**: Quotation file + requirement description.
- **Action**: Extract raw text and structured fields (unit price, currency, lead time, terms).
- **Output**: Normalized quote record in the workspace.

3.2 Stage 2: Research Planning

The reasoning model decomposes the task into research steps.

- **Input**: Normalized quote + requirements.
- **Action**: Identify which suppliers to research, what specifications to verify, and which sources to consult.

- **Output**: A research plan stored in the workspace.

3.3 Stage 3: Web Research

The browsing layer executes the plan against the open web.

- **Input**: Research plan.
- **Action**: Multi-engine search, fetch supplier pages, catalogs, and reviews. Escalate to anti-detect browser only when blocked.
- **Output**: Raw retrieved content converted to Markdown, stored per supplier.

3.4 Stage 4: Evidence Storage

Findings are written to a file-based workspace rather than held in context.

- **Input**: Retrieved content from Stage 3.
- **Action**: Save each finding as a Markdown file tagged by supplier and topic.
- **Output**: Persistent, retrievable evidence store.

3.5 Stage 5: Comparison and Reasoning

The reasoning model analyzes the stored evidence against the quote.

- **Input**: Quote record + evidence store.
- **Action**: Compare suppliers on specifications, price, lead time, and reputation. Determine ranking and justification.
- **Output**: Draft recommendation with supporting reasoning.

3.6 Stage 6: Citation Verification

Every claim in the draft is checked against its source.

- **Input**: Draft recommendation.
- **Action**: Trace each factual claim to a retrieved source. Flag or remove unsupported statements.
- **Output**: Verified recommendation with inline citations.

3.7 Stage 7: Final Output

The verified result is returned to the user.

- **Input**: Verified recommendation.
- **Action**: Format as ranked supplier list with reasoning and citation links.
- **Output**: Ranked suppliers + justification + source citations + confidence notes.

3.8 Iteration

Stages 3–6 may repeat. If verification flags a claim as unsupported, the agent can return to Stage 3 to gather additional evidence before finalizing. This loop continues until citation accuracy meets the threshold defined in the evaluation section.

4 Literature Summary

4.1 Reasoning Formats and GSM8K Performance

The reference paper compares deep reasoning formats on mathematical reasoning benchmarks. Table 2 reproduces the reported scores. The paper does **not** include DeepSeek V3.1 (released later); it covers earlier reasoning models and reasoning-format ablations.

Table 2: Performance of reasoning formats, sorted primarily by GSM8K score.

Model	Base Model	GSM8K	MATH	GPQA	Olympiad	LiveCode
Latent space deep reasoning						
No-CoT	Mistral-7B	38.0	–	–	–	–
SQ-VAE	Llama-2-7B	40.0	7.0	–	–	–
RecurrentBlock-3.5B	–	42.1	–	–	–	–
ICoT-SI	Mistral-7B	51.0	–	–	–	–
Natural language deep reasoning						
Self-Rewarding	Llama-2-7B	40.0	10.7	–	–	–
Llama-3.1-8B-DAR	–	56.7	20.3	–	–	–
MetaMath	Llama-2-7B	66.5	–	–	–	–
OVM	Llama-2-7B	73.7	–	–	–	–
NuminaMath-7B-CoT	–	75.4	55.2	–	19.9	–
Qwen2-7B	–	79.9	44.2	–	21.3	–
Qwen2-Math-7B	–	80.4	50.4	–	38.2	–
Internlm2-math-plus-7B	–	84.0	54.4	–	18.8	–
OMI2	Qwen2.5-Coder-7B	84.1	72.3	36.2	–	27.2
Llama-3.1-70B	–	85.5	41.4	–	–	–
CODEI-O++	Qwen2.5-Coder-7B	85.7	72.1	40.6	–	29.1
CODEI-O	Qwen2.5-Coder-7B	86.4	71.4	43.3	–	28.5
W1	Qwen2.5-Coder-7B	87.0	71.4	39.1	–	26.0
W1 (Full)	Qwen2.5-Coder-7B	87.0	71.1	42.9	–	27.6
OMI2 (Full)	Qwen2.5-Coder-7B	88.5	73.2	40.9	–	28.4
DeepSeek-Math-7B-RL	–	88.2	51.7	–	19.0	–
Llama-3.1-405B	–	89.0	53.8	–	–	–
CoMAT	GPT-4	93.7	–	40.4	–	–
CoT	GPT-4	94.5	–	41.8	50.2	–
FCoT	GPT-4	95.0	–	–	–	–
Qwen2.5-Math-7B-Instruct	GPT-4	95.2	83.6	–	41.6	–
MathPrompt	GPT-4	95.6	–	–	–	–
Qwen2.5-Math-7B-Instruct	–	95.9	85.9	–	49.0	–
DeepSeek-R1-Distill-Qwen-7B	–	–	92.8	–	49.1	37.6
DeepSeek-R1-Distill-Qwen-32B	–	–	94.3	–	62.1	57.2
Structured language deep reasoning						
STaR	Llama-2-7B	58.2	16.0	–	–	–
ENVISIONS	Llama-2-7B	59.0	19.0	–	–	–
MaMA-TH	Code-Llama-7B	59.4	–	–	–	–
MathCoder-CL	Code-Llama-7B	67.8	30.2	–	–	–

Model	Base Model	GSM8K	MATH	GPQA	Olympiad	LiveCode
ToRA-Code	Llama-2-7B	72.6	–	–	–	–
Brain	Code-Llama-7B	74.0	–	–	–	–
DeepSeek-Coder-7B	–	77.4	44.4	–	–	–
Smaug	Qwen2-Math-Base	81.5	50.0	–	–	–
OC-SFT-1.34	Qwen2.5-Coder-7B	86.7	70.9	37.7	–	27.5
PyEdu	Qwen2.5-Coder-7B	85.8	71.4	40.9	–	25.8
Qwen2.5-Math-7B-Instruct	–	94.6	85.2	–	55.6	–
Qwen2.5-Math-72B-Instruct	–	95.8	88.1	–	60.6	–
QuaSAR	GPT-4	96.5	–	55.4	44.6	–
MathDivide	GPT-4	96.8	–	–	–	–

4.2 Key Observations from the Paper

- **Natural language CoT dominates**: The highest GSM8K scores come from natural-language reasoning with strong base models (Qwen2.5-Math, GPT-4).
- **Latent-space reasoning underperforms**: Recurrent and SQ-VAE methods lag far behind explicit CoT.
- **Structured reasoning (STaR, ToRA, Smaug)** closes the gap but does not surpass natural-language CoT at the top end.
- **GPT-4 baselines remain strong**, but open models (Qwen2.5-Math-72B at 95.8) are now competitive.
- The paper does **not** benchmark agentic tool use, web browsing, or citation accuracy—the metrics most relevant to this project.

5 Proposed Models

5.1 Reasoning Model Candidates

Model	Size	License	Notes
DeepSeek V3.1	671B total (37B active), MoE	MIT	Agent-optimized, hybrid thinking mode, search-tool format. **Primary choice.**
Tongyi DeepResearch	Not specified	Open	58.3% BrowseComp, 38.3% HLE. Strong research baseline.
Qwen3 Series	Varies (7B–235B)	Custom	100M MAU threshold requires separate license beyond that.
Kimi K2 Thinking	Large MoE	Modified MIT	100M MAU / $20M revenue threshold.

Table 3: Reasoning model candidates for enterprise use.

5.2 DeepSeek V3.1 Size Reference

Deployment note: DeepSeek V3.1 cannot run on Google Colab's free T4 GPU (15–16 GB VRAM). Even the smallest quantized build (170 GB) is more than ten times the available memory. Colab prototyping must use a smaller model (7B–14B) or an API endpoint.

Format	Approximate Size
Full (FP8 / original)	∼689 GB
Full (disk requirement)	∼715 GB
Quantized 4-bit	∼384 GB
Quantized 1-bit	∼170 GB

Table 4: DeepSeek V3.1 storage requirements by quantization.

Model	Params	License	Notes
Qwen2.5-7B-Instruct	7B	Qwen (threshold)	Native tool calls, fits T4 in 4-bit.
LFM2.5-2.6B	2.6B	Open	Lightweight, OpenAI-style tool calls.
Nex-N2.5-Mini	Small	Open	Reasoning-effort parameter, agentic tuning.
Llama-3.1-8B-Instruct	8B	Llama Community	Widely supported, tool-calling capable.

Table 5: Smaller candidates suitable for Colab-based proof of concept.

5.3 Smaller Models for Colab Prototyping

6 Browsing Layer

6.1 Candidate Tools

Tool	Anti-Blocking Features	License	Limitations
Hound	Real Chrome TLS, 4 fingerprint profiles, Bezier mouse curves, multi-engine fallback	MIT	Does not bypass DataDome, Akamai, or Cloudflare Turnstile.
GhostFetch	Ghost Protocol fingerprinting, proxy rotation, smart scrolling	MIT	No interactive CAPTCHA bypass; login sites out of scope.
Penumbra	Tor routing with fresh circuits per source	Open	May trigger additional verification on some sites.

Table 6: Browsing layer candidates.

6.2 Blocking Reality

Both Hound and GhostFetch **reduce** but do not **eliminate** blocking. Modern enterprise anti-bot systems (Cloudflare Precursor, DataDome, Akamai) perform continuous behavioral analysis and cross-layer signature consistency checks. Key implications:

- Agent must handle `403`, `429`, and CAPTCHA responses by switching sources, not retrying blindly.
- Residential proxy rotation is required for scale.
- Respect `robots.txt` and site terms of service.

7 Summary

- **Primary reasoning model**: DeepSeek V3.1 (MIT, agent-optimized, 671B MoE).
- **Browsing layer**: Hound (MIT, self-hosted, multi-engine, auto-escalation).

Risk	Mitigation
Model hallucinates supplier facts	Citation verification layer blocks unsupported claims.
Websites block the agent	Multi-engine fallback, proxy rotation, graceful degradation.
Context overflow on long research	File-based workspace with on-demand retrieval.
License violation at scale	Pre-cleared MIT components; avoid threshold-licensed models unless cleared.
High per-analysis cost	Benchmark token usage; cache supplier data.

Table 7: Risks and mitigations.

- **Memory**: File-based workspace to survive long research trajectories.
- **Verification**: Custom citation checker to eliminate unsupported claims.
- **Evaluation**: Standardized prompt across candidate models with citation-accuracy target above 95%.
- **Key constraint**: DeepSeek V3.1 cannot run on Colab free tier; prototype with 7B–14B models first.

Appendix: Referenced Sources

- Reference paper: deep reasoning format comparison (GSM8K benchmark table reproduced in Section 4.1).
- DeepSeek V3.1 model card and license (MIT).
- Hound MCP server documentation and limitations.
- GhostFetch documentation and limitations.
- Tongyi DeepResearch benchmark results (BrowseComp 58.3%, HLE 38.3%).